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Business Context:

Employee Promotion means the ascension of an employee to higher ranks — the ultimate reward for dedication and loyalty towards an organization. The HR team plays a critical role in handling all promotion tasks based on ratings and other performance attributes.

The HR team in JMD Company stored data from the promotion cycle last year, which consists of details of all employees who worked during that year and whether they got promoted or not. However, every time this process gets delayed because comparing so many employee attributes manually make it difficult to reach consistent and timely decisions.

Problem Definition:

For the upcoming appraisal cycle, the HR team wants to leverage the stored data and machine learning to build a model that predicts whether a person is eligible for promotion or not. The goal is to:

1. Explore and visualize the dataset to uncover meaningful patterns
2. Build a classification model to predict promotion eligibility
3. Optimize the model using appropriate techniques
4. Generate actionable insights and recommendations for the HR team

Data Background:

The dataset contains details of all employees at JMD Company for the previous promotion cycle, with the target variable indicating whether each employee was recommended for promotion.

Column Name	Description
employee_id	Unique ID for the employee
department	Department of employee
region	Region of employment (unordered)
education	Education level of employee
gender	Gender of employee
recruitment_channel	Channel of recruitment for employee
no_of_trainings	No. of other trainings completed in the previous year on soft/technical skills
age	Age of employee
previous_year_rating	Employee rating for the previous year (1–5 scale)
length_of_service	Length of service in years
awards_won	1 if awards won during previous year, else 0

Column Name	Description
avg_training_score	Average score in current training evaluations (39–99)
is_promoted	(Target) Recommended for promotion: 1 = Yes, 0 = No

Data Overview:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 54808 entries, 0 to 54807
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   employee_id            54808 non-null  int64
1   department             54808 non-null  object
2   region                 54808 non-null  object
3   education              52399 non-null  object
4   gender                 54808 non-null  object
5   recruitment_channel    54808 non-null  object
6   no_of_trainings        54808 non-null  int64
7   age                   54808 non-null  int64
8   previous_year_rating   50684 non-null  float64
9   length_of_service      54808 non-null  int64
10  awards_won             54808 non-null  int64
11  avg_training_score     52248 non-null  float64
12  is_promoted            54808 non-null  int64
dtypes: float64(2), int64(6), object(5)
```

Columns with missing values:

	Missing Count	Missing %
previous_year_rating	4124	7.524
avg_training_score	2560	4.671
education	2409	4.395

- The dataset contains 54,808 rows and 13 columns (12 after dropping employee_id).
- No duplicate records are present, which confirms data integrity at the row level.
- Three columns have missing values: education (~4.4%), previous_year_rating (~7.5%), and avg_training_score (~4.7%). These will be imputed during preprocessing.
- The target variable is_promoted is heavily imbalanced: approximately 91.5% of employees were not promoted (0) versus only 8.5% who were promoted (1) — a ratio of roughly 10.75:1.
- Categorical columns include department (9 unique), region (34 unique), education (3 unique), gender (2 unique), and recruitment_channel (3 unique).
- avg_training_score ranges from 39 to 99 with a mean of ~63.7; age ranges from ~20 to ~60 years; length_of_service ranges from 1 to ~37 years, indicating a wide spread of employee tenures.

Exploratory Data Analysis:

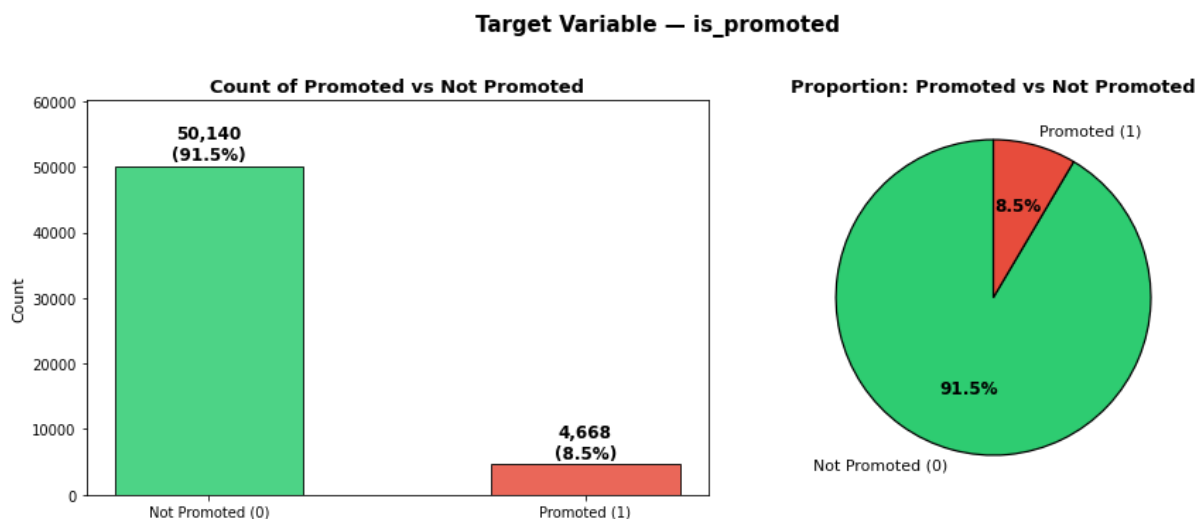
Each variable is visualized using the most appropriate chart type:

- Continuous numeric variables: Combined boxplot + histogram with KDE to capture distribution shape, spread, central tendency, and outliers
- Discrete / categorical variables: Annotated bar charts with count and percentage labels
- Bivariate analysis (vs. target): Grouped bar charts, stacked bar charts, and side-by-side distribution plots to reveal how each feature relates to promotion outcome

Univariate Analysis:

Each variable is individually analysed to understand its distribution, central tendency, spread, and skewness before examining relationships with the target variable.

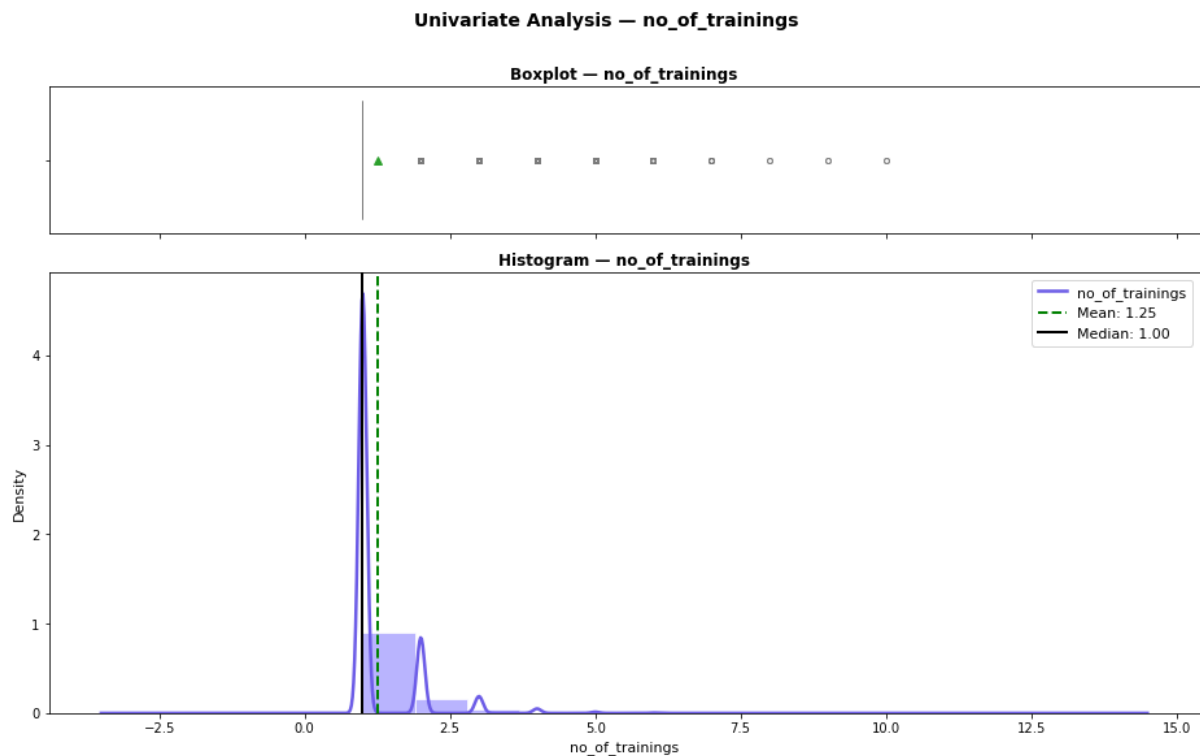
1. Target Variable — is_promoted



[Figure: Target Variable Distribution — Bar Chart and Pie Chart]

- The target variable is severely imbalanced: approximately 91.5% of employees were not promoted while only 8.5% were recommended for promotion — a ratio of roughly 10.75:1.
- This extreme imbalance means that a classifier predicting 'Not Promoted' for everyone would achieve ~91.5% accuracy, making raw accuracy a misleading metric. The evaluation strategy must therefore prioritize F1 Score (macro), which balances precision and recall across both classes.
- The imbalance also necessitates the use of resampling strategies (SMOTE oversampling and random undersampling) during model building to prevent the classifier from becoming biased towards the majority class.

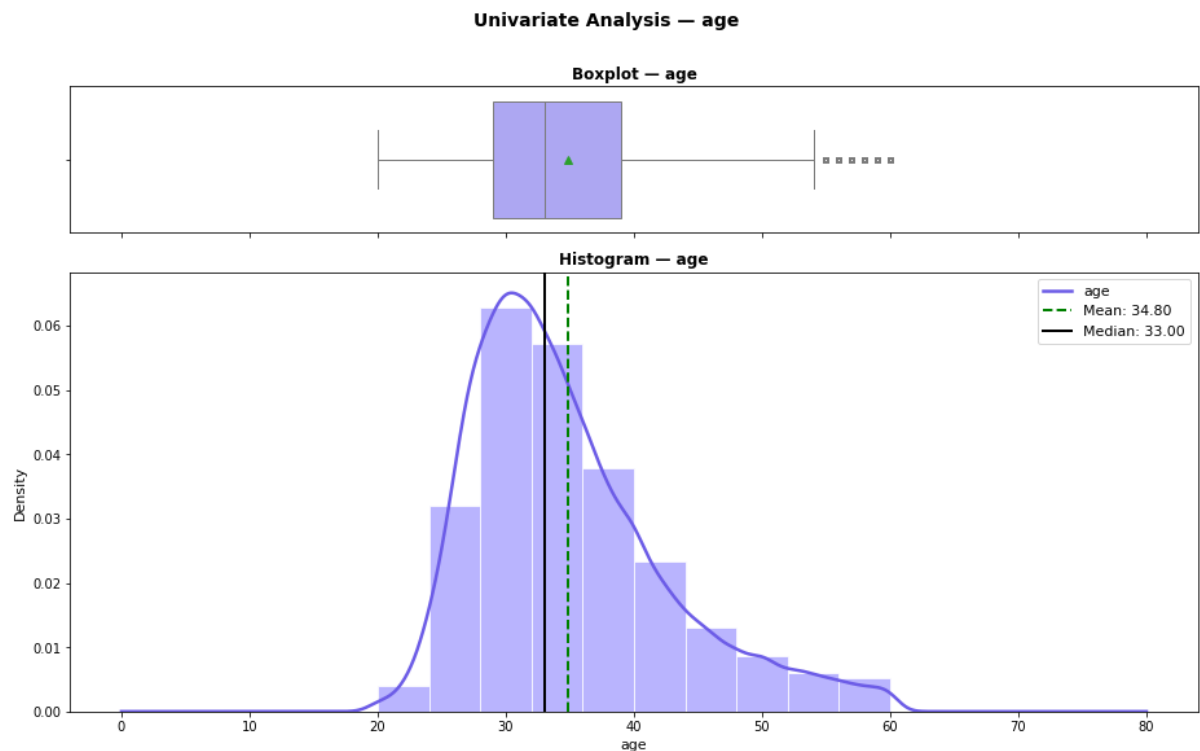
2. No. of Trainings



[Figure: Univariate Analysis — no_of_trainings (Boxplot + Histogram)]

- The distribution is heavily right-skewed: the vast majority of employees completed only 1 training in the previous year, with count dropping sharply for 2 or more trainings. The maximum is 10 trainings.
- Mean (~1.25) is slightly higher than the mode (1), driven by the right tail. The extreme values (>5 trainings) are likely outliers or represent particularly active employees.
- The IQR is very narrow ($Q1 = Q3 = 1$), confirming that most employees cluster at the minimum number of trainings, which limits the variable's discriminatory power on its own.

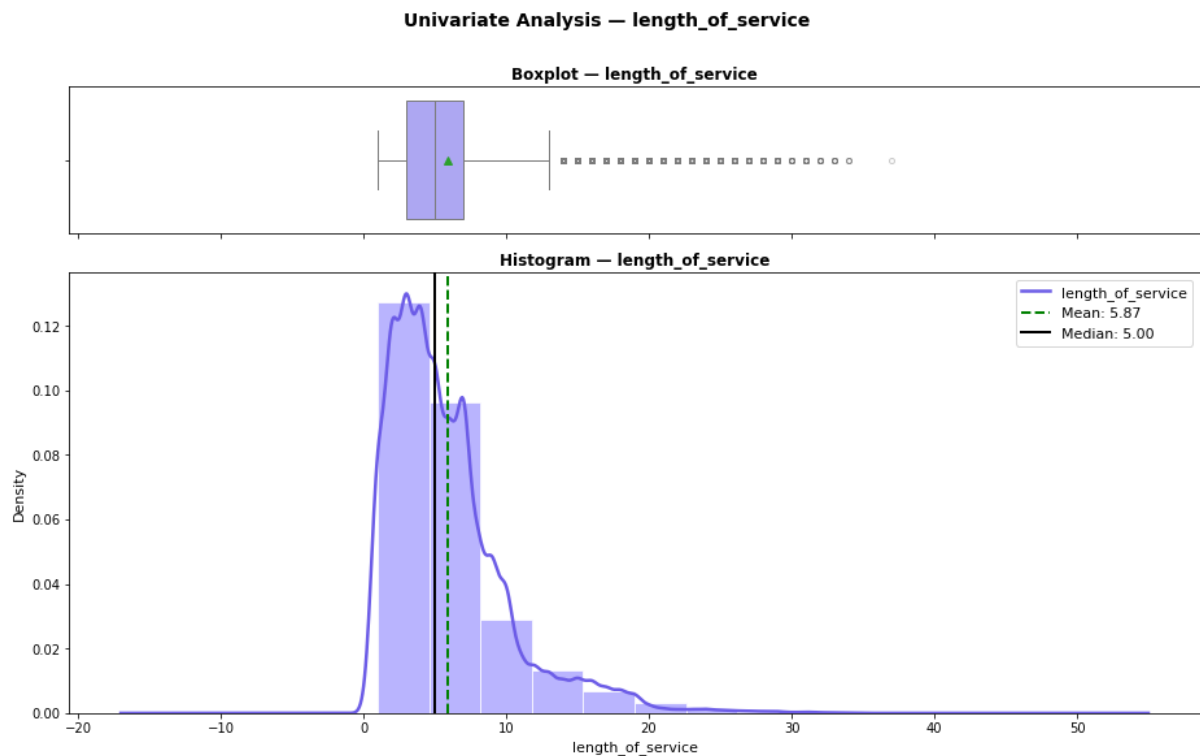
3. Age of Employee



[Figure: Univariate Analysis — age (Boxplot + Histogram)]

- The distribution of age is roughly **right-skewed**, with the mean (~34.8 years) slightly above the median. Most of the employees fall in the **25–40 age bracket**, consistent with an organization's core working-age population.
- There are some right-tail outliers representing older employees (above ~55 years), visible in the boxplot.
- The wide spread suggests a diverse workforce, and age may correlate with experience and seniority — making it a potentially informative predictor of promotion eligibility.

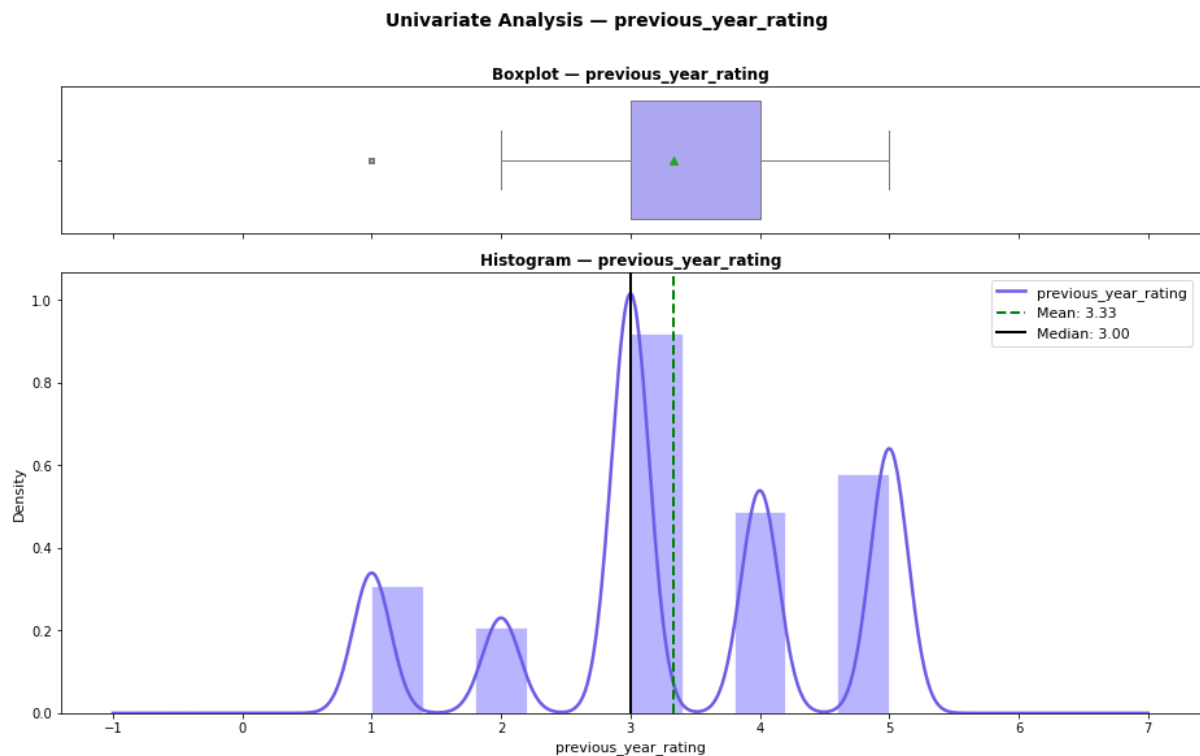
4. Length of Service



[Figure: Univariate Analysis — length_of_service (Boxplot + Histogram)]

- length_of_service is right-skewed, with a peak at 2–5 years and a long tail extending to ~37 years. The median is around 5–6 years.
- The presence of employees with very long tenures (20+ years) introduces outliers. However, these are genuine data points and should not be removed as longer-tenured employees may have different promotion dynamics.
- High tenure may reflect promotability through experience, or alternatively, stagnation — making this an interesting variable to examine in bivariate analysis.

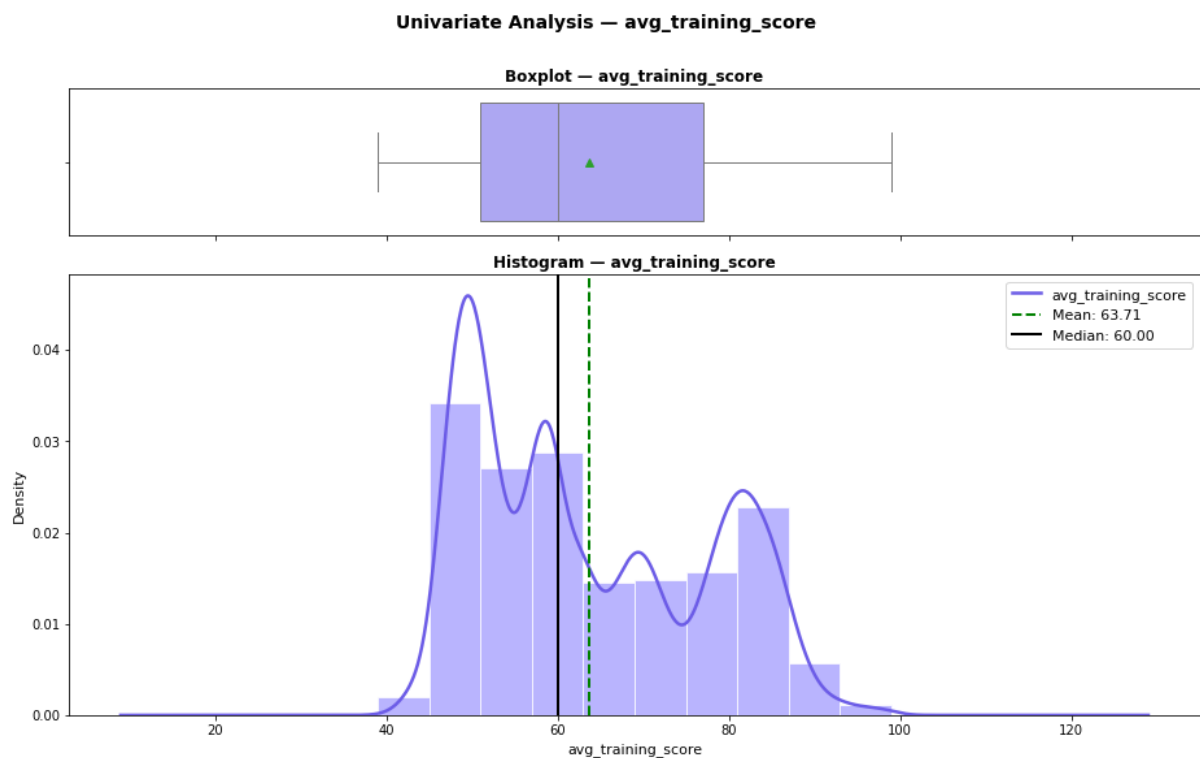
5. Previous Year Rating



[Figure: Univariate Analysis — previous_year_rating (Boxplot + Histogram)]

- Ratings are discrete (1–5 scale) rather than continuous, making the distribution bar-like. The most common rating is 3, followed by 4 and 5.
- Ratings of 1 and 2 are relatively rare, indicating that the majority of employees received a satisfactory or above-satisfactory review in the previous year.
- Since this is a performance metric directly evaluated by managers, it is expected to be a strong predictor of promotion decisions.

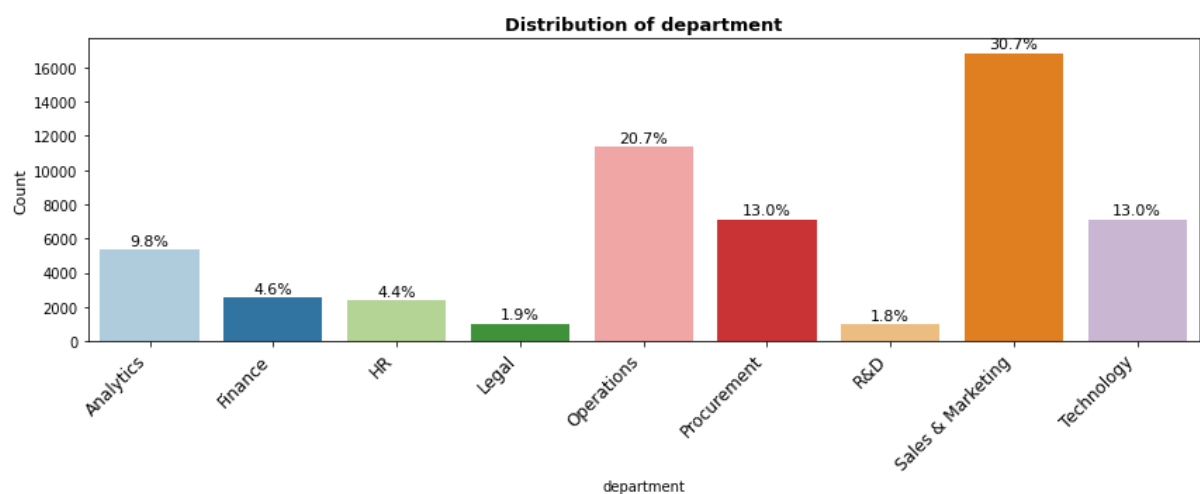
6. Average Training Score



[Figure: Univariate Analysis — avg_training_score (Boxplot + Histogram)]

- avg_training_score ranges from 39 to 99 with a roughly bimodal-like distribution: a cluster in the 40–60 range and another cluster in the 70–90 range.
- The mean (~63.7) and median (~60) are moderately aligned, suggesting only mild skew. There are no extreme outliers.
- Given that training scores directly reflect skill development and engagement, this feature is expected to be one of the most discriminating variables for predicting promotion — employees who score high are likely more capable and visible to management.

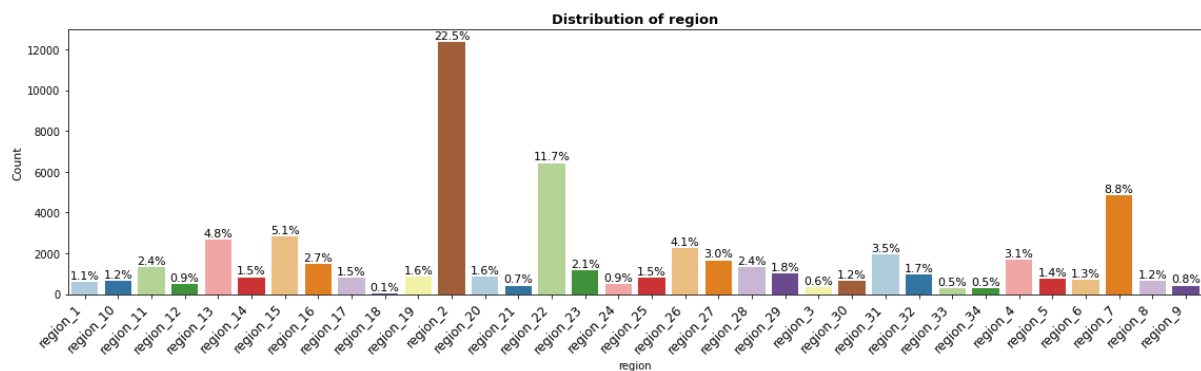
7. Department



[Figure: Distribution of department (Annotated Bar Chart)]

- Sales & Marketing and Operations are the two largest departments, together accounting for over 50% of the workforce.
- Smaller departments include Legal, R&D, and HR, which are specialized functions with fewer headcount requirements.
- Departmental distribution matters because promotion rates likely vary by department — departments with more structured career paths (e.g., Finance, Technology) may have different promotion dynamics compared to support functions.

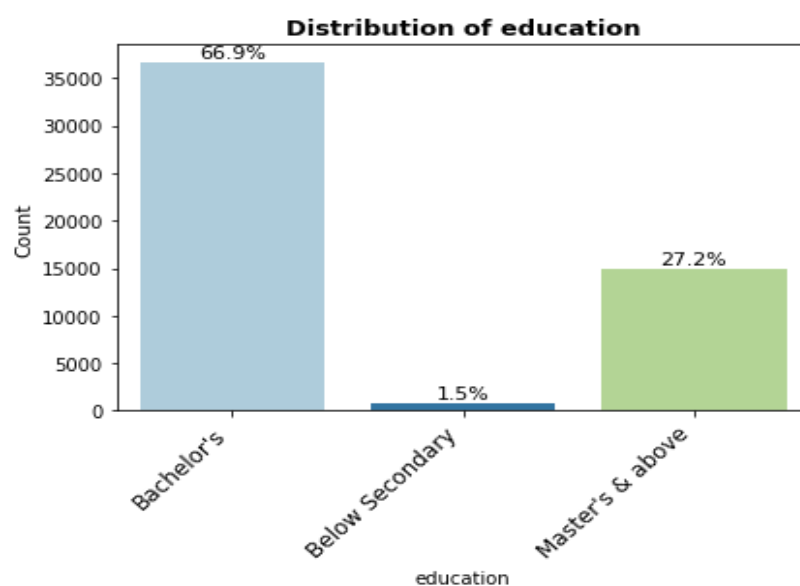
8. Region



[Figure: Distribution of region (Annotated Bar Chart)]

- There are 34 regions, making this a high-cardinality categorical variable. Region 2 has the highest employee count, while several other regions have relatively few employees.
- The high cardinality means that one-hot encoding will create 33 dummy variables, increasing dimensionality. Tree-based models handle this well, but it may increase risk of over-fitting.
- The anonymized nature of region labels (region_1 to region_34) limits direct business interpretation.

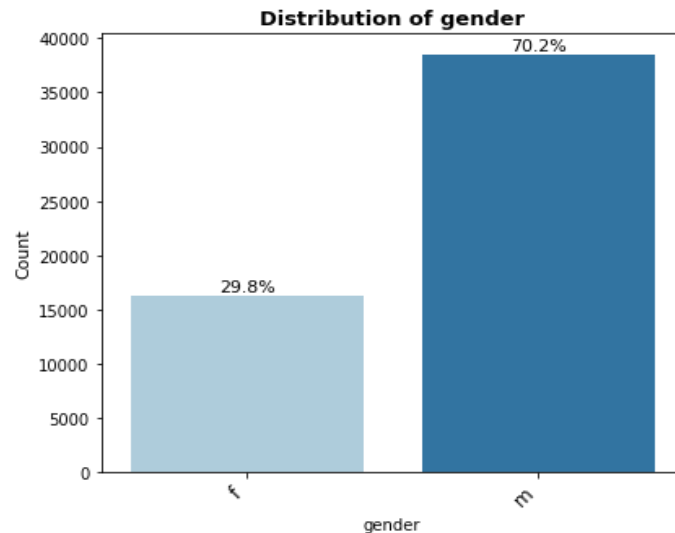
9. Education Level



[Figure: Distribution of education (Annotated Bar Chart)]

- Bachelor's degree holders form the largest group (~71%), followed by Master's & above (~22%) and Below Secondary (~7%).
- The education column also has ~4.4% missing values, which will be imputed using the most frequent value (Bachelor's) during preprocessing.
- Higher education levels may correlate with eligibility for senior promotions, making this a relevant feature to investigate in bivariate analysis.

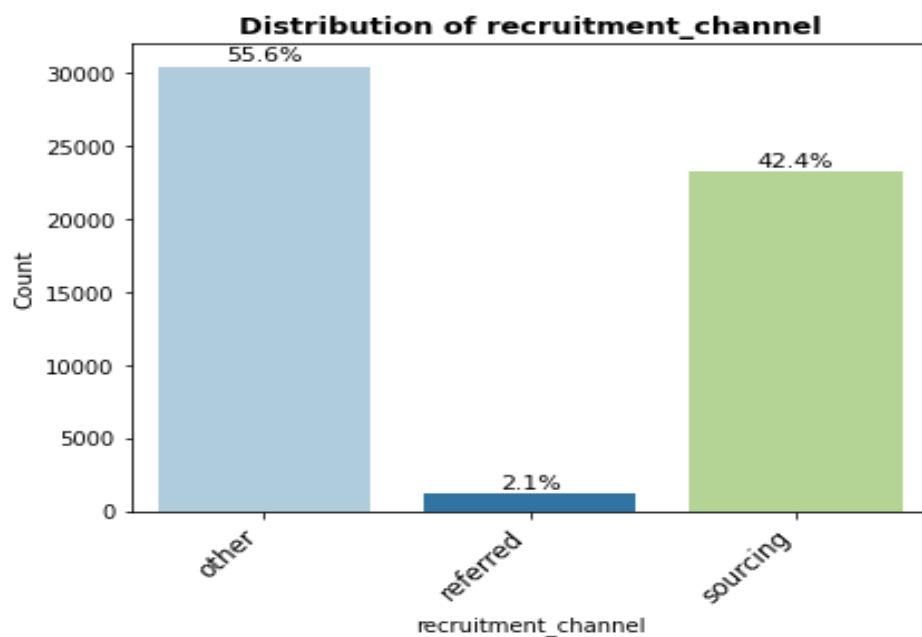
10. Gender



[Figure: Distribution of gender (Annotated Bar Chart)]

- The workforce is predominantly male (~59%) vs. female (~41%). While not a severe gender imbalance, it is notable.
- Gender is expected to have limited direct predictive power for promotion in an ideal meritocratic system, but bivariate analysis will reveal if historical promotion patterns show any disparity.

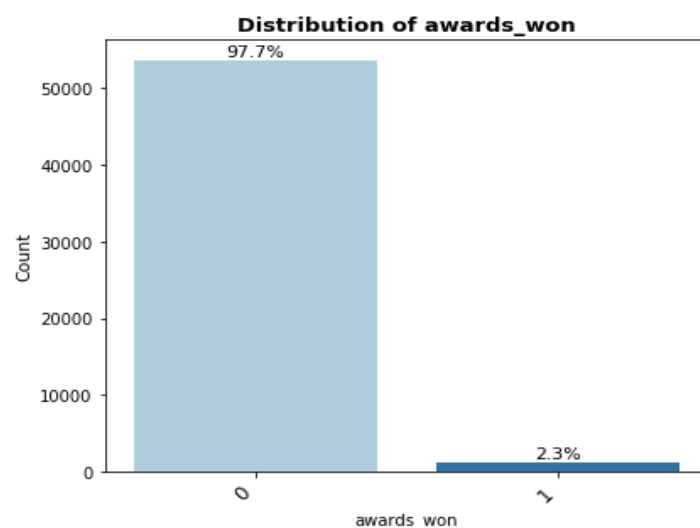
11. Recruitment Channel



[Figure: Distribution of recruitment_channel (Annotated Bar Chart)]

- Sourcing (direct/proactive recruitment) is the dominant channel (~54%), followed by Other (~41%) and Referred (~5%).
- Referred employees tend to come in through trusted networks and may perform differently than sourcing hires — an interesting angle to explore in the bivariate section.

12. Awards Won



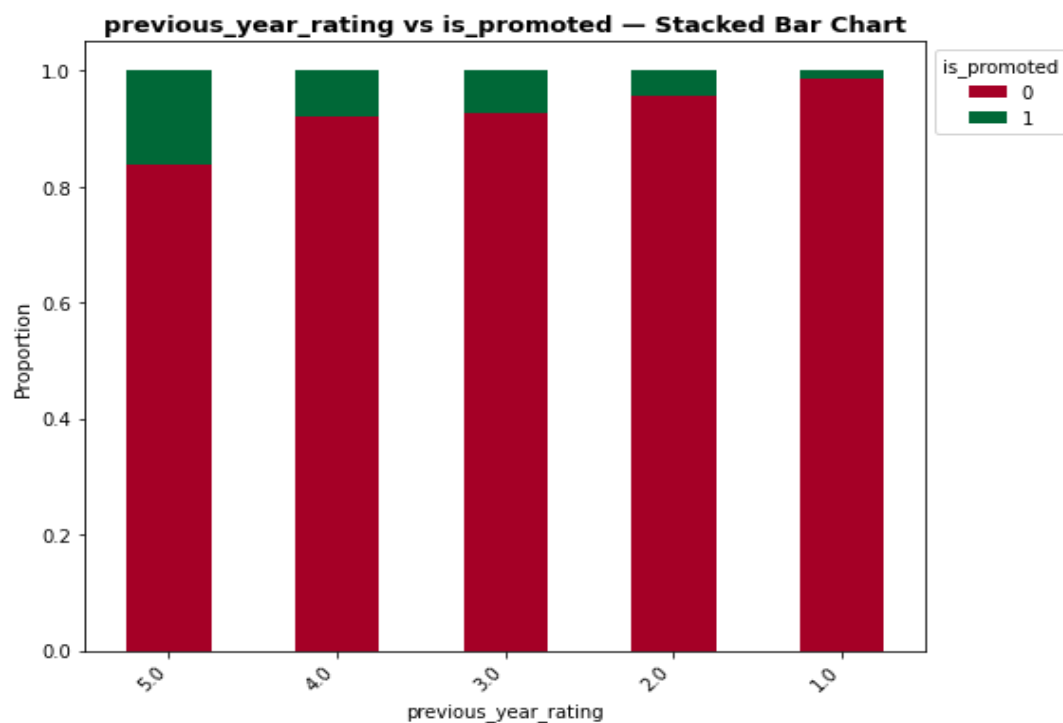
[Figure: Distribution of awards_won (Annotated Bar Chart)]

- The vast majority (~96%) of employees did not win an award in the previous year, while only ~4% did.
- Despite low prevalence, award winners are likely high performers and this binary variable may be a strong signal for promotion eligibility.

Bivariate Analysis:

Each predictor is analysed against the target variable `is_promoted` to identify factors that differentiate promotion outcomes. Stacked bar charts are used for categorical predictors; distribution plots and boxplots are used for numerical predictors.

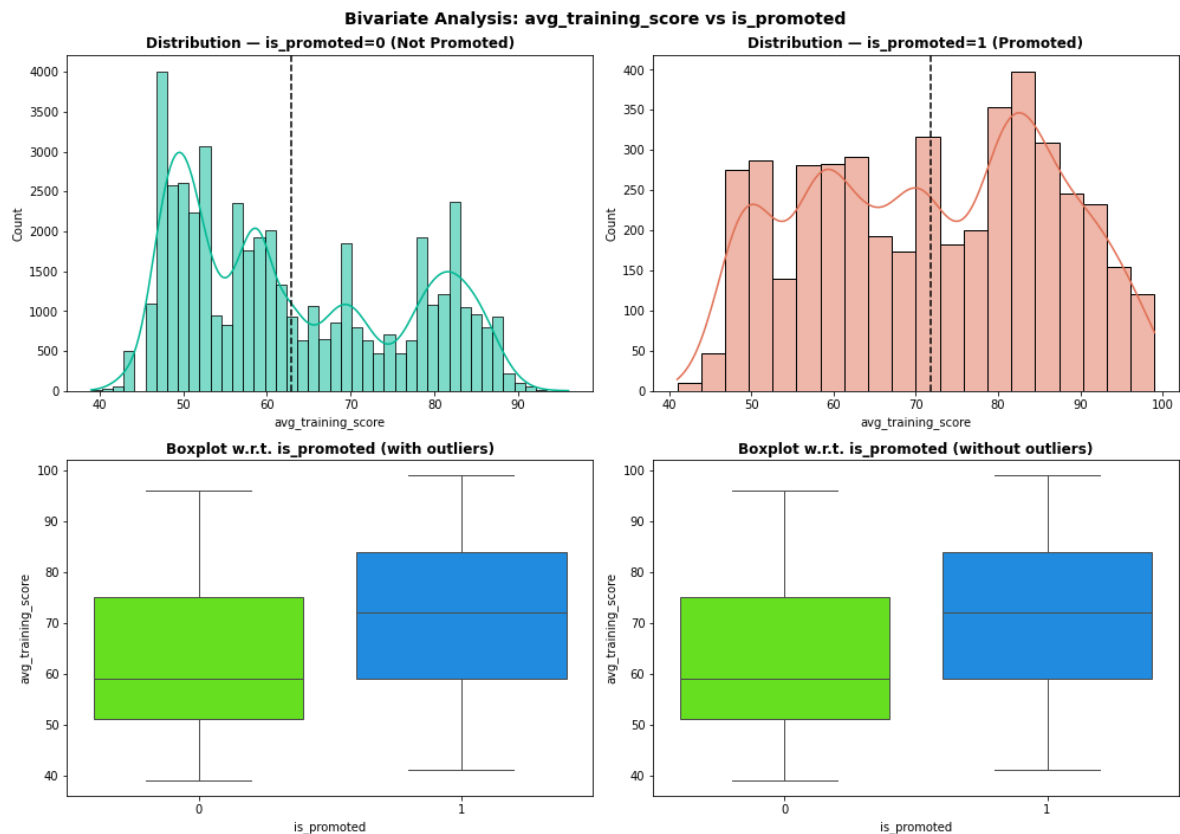
1. `previous_year_rating` vs. `is_promoted`



[Figure: `previous_year_rating` vs `is_promoted` — Stacked Bar Chart]

- There is a clear positive relationship between previous year rating and promotion probability: employees with a rating of 5 have the highest promotion rate (~19%), while those with rating 1 have the lowest (<1%).
- Employees with ratings of 4 and 5 account for a disproportionate share of all promotions, making this variable one of the strongest predictors of promotion eligibility.
- This aligns with business expectations — consistently high performers are more likely to be recognized and promoted.

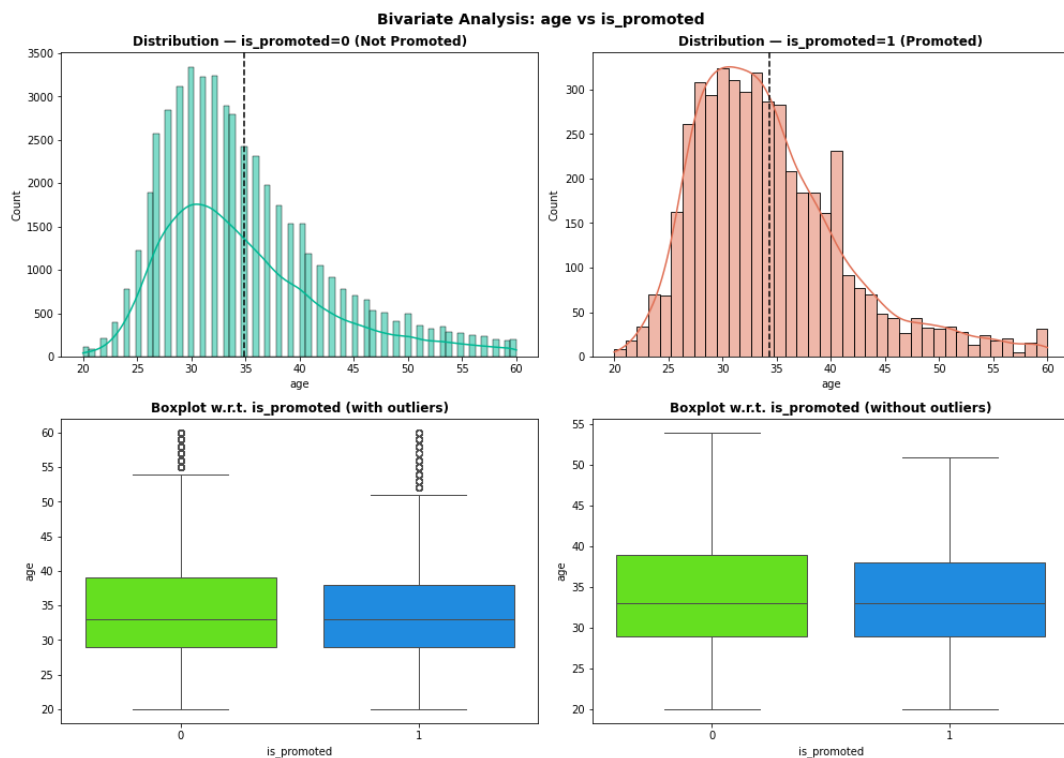
2. avg_training_score vs. is_promoted



[Figure: avg_training_score vs is_promoted — Distribution Plots and Boxplots]

- Promoted employees have a noticeably higher average training score distribution compared to non-promoted employees. The median training score for promoted employees is substantially higher (~80–85) versus non-promoted (~55–60).
- The bimodal pattern is even more visible when segmented by promotion status, with promoted employees dominating the upper mode (70–90 range).
- avg_training_score is likely to be the single most important feature in the final model — a direct reflection of employee capability and organizational visibility.

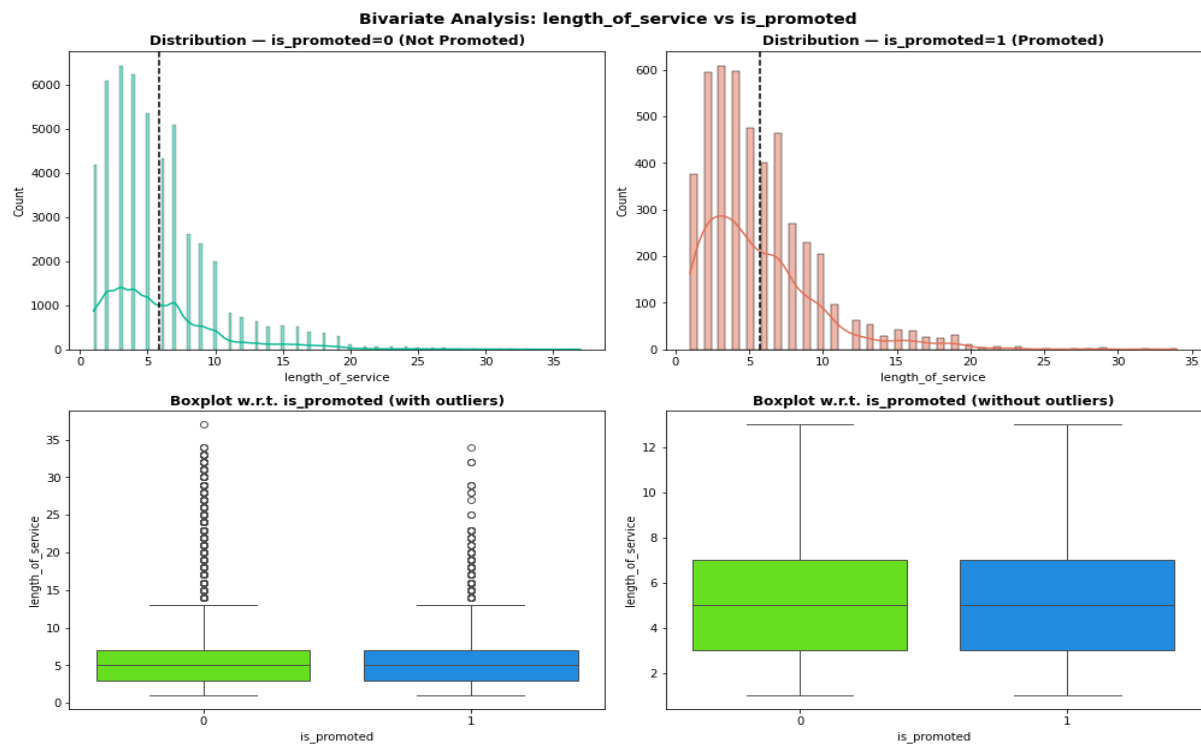
3. age vs. is_promoted



[Figure: age vs is_promoted — Distribution Plots and Boxplots]

- Promoted employees tend to be slightly older (median ~35–37 years) compared to non-promoted employees (~33–35 years). However, the difference is modest and the distributions overlap heavily.
- Younger employees (<25) are rarely promoted, likely because they have insufficient tenure and performance history. The promotion sweet spot appears to be the 32–42 age range.
- Age alone is not a strong discriminator — its predictive power likely comes through interaction with length_of_service and previous_year_rating.

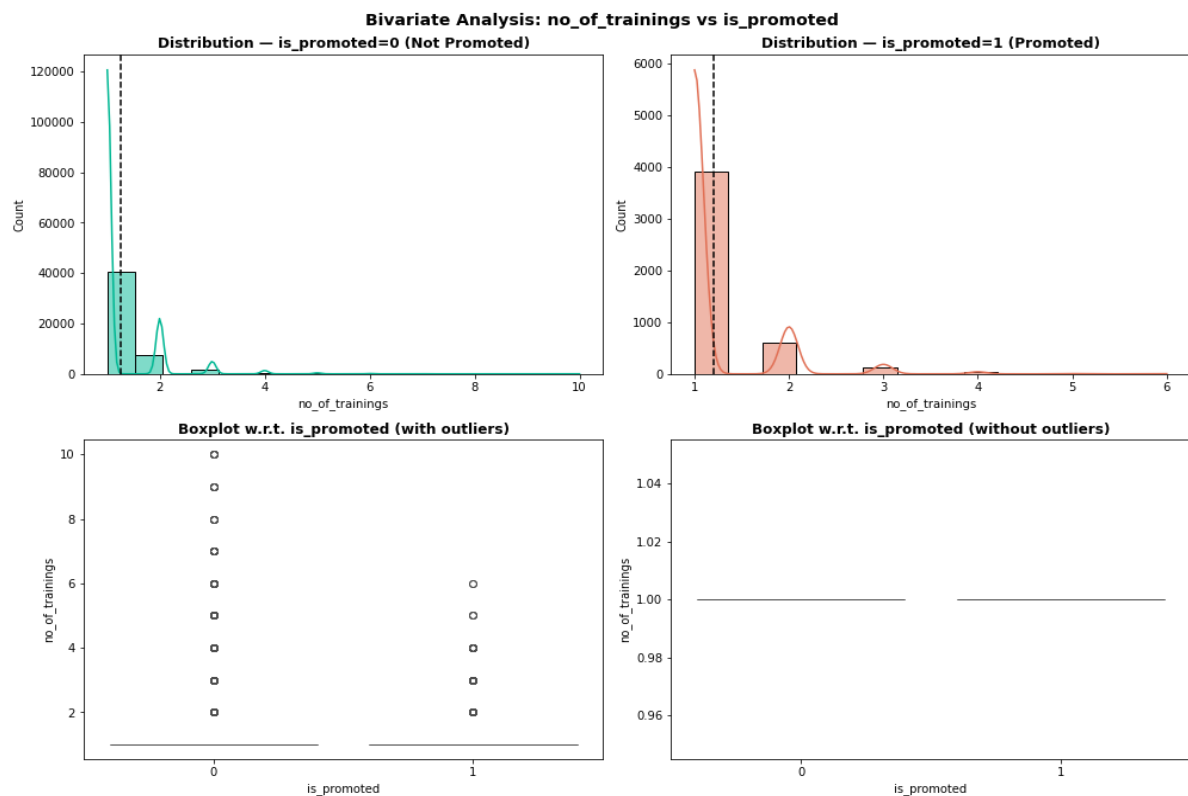
4. length_of_service vs. is_promoted



[Figure: length_of_service vs is_promoted — Distribution Plots and Boxplots]

- Promoted employees generally have moderate to high service tenure; the distribution for promoted employees' peaks at 5–10 years.
- Very short-tenured employees (<2 years) are rarely promoted, which makes intuitive sense as they haven't had sufficient time to demonstrate capability.
- Extremely long-tenured employees (20+ years) also show a relatively lower promotion rate, possibly indicating that promotions tend to happen at earlier career stages.

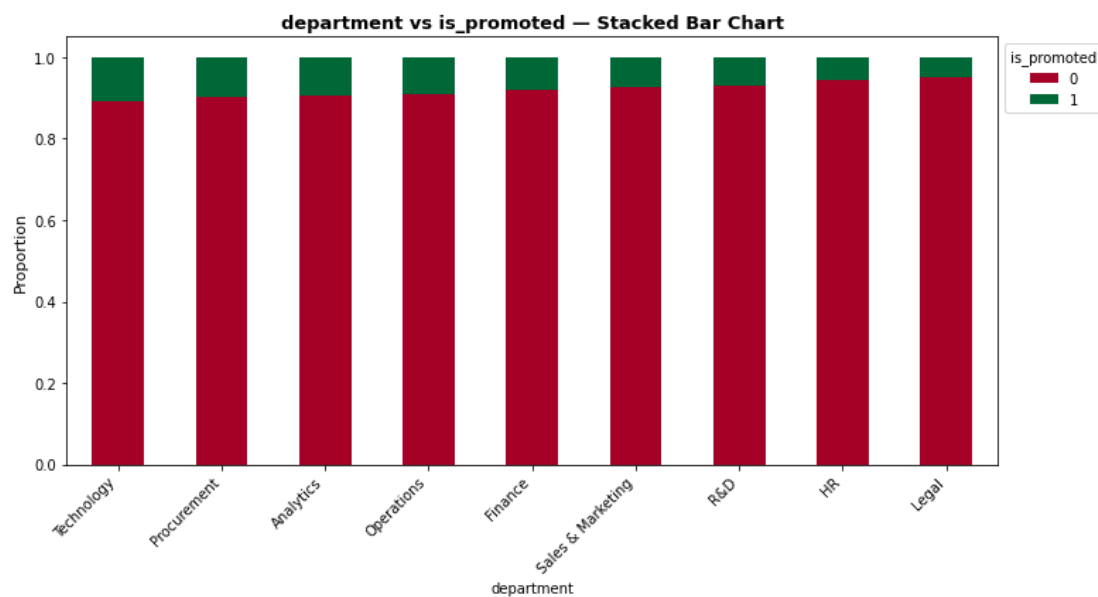
5. no_of_trainings vs. is_promoted



[Figure: no_of_trainings vs is_promoted — Distribution Plots and Boxplots]

- Counterintuitively, employees who completed only 1 training actually have a slightly higher raw count of promotions — but this is purely because most employees completed just 1 training .
- When looking at the proportion within each training count group, there is no strong monotonic pattern: completing more trainings does not reliably increase promotion probability, suggesting the quality of training (captured by avg_training_score) matters far more than quantity.

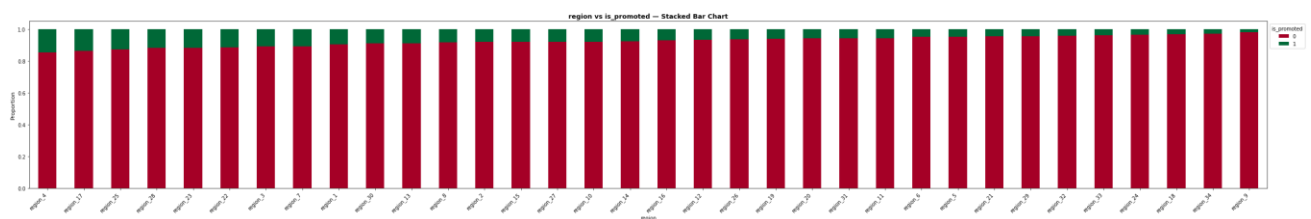
6. department vs. is_promoted



[Figure: department vs is_promoted — Stacked Bar Chart]

- Sales & Marketing has the highest absolute number of promotions (by volume), largely because it is the largest department. However, in terms of promotion rate, Technology and Analytics departments tend to have higher proportional promotion rates.
- HR and Legal have among the lowest promotion rates, reflecting their smaller size and possibly less hierarchical advancement opportunities.
- This department-level variation is important for the HR team — it suggests that promotion benchmarks may need to be department-specific rather than organization-wide.

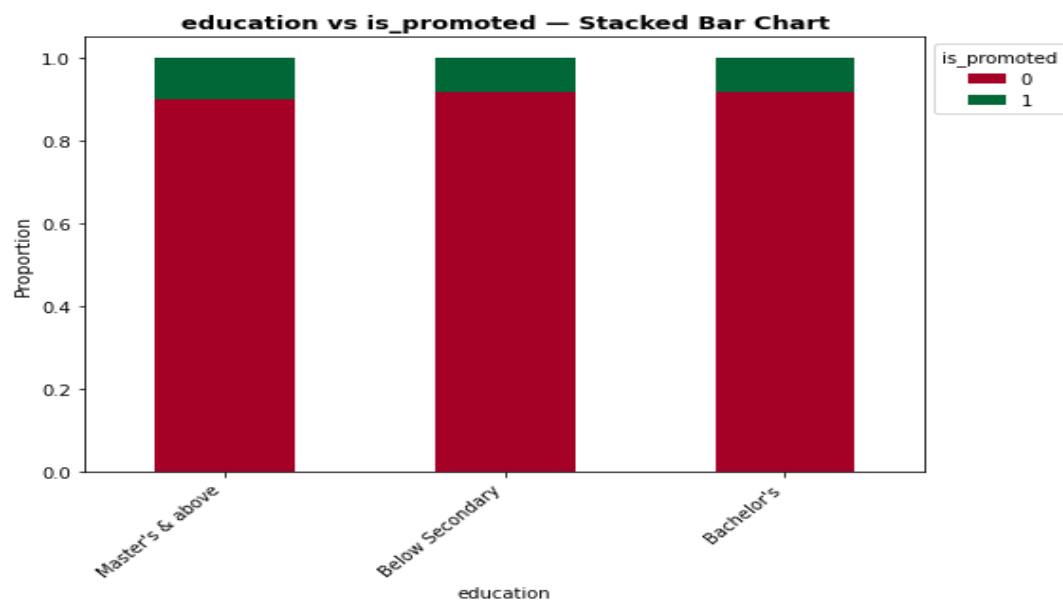
7. region vs. is_promoted



[Figure: region vs is_promoted — Stacked Bar Chart]

- Promotion rates vary across regions, but given the large number of categories (34), it is difficult to draw individual region-level conclusions without further aggregation.
- Some regions have notably higher promotion proportions, while others are closer to the overall base rate of ~8.5%.

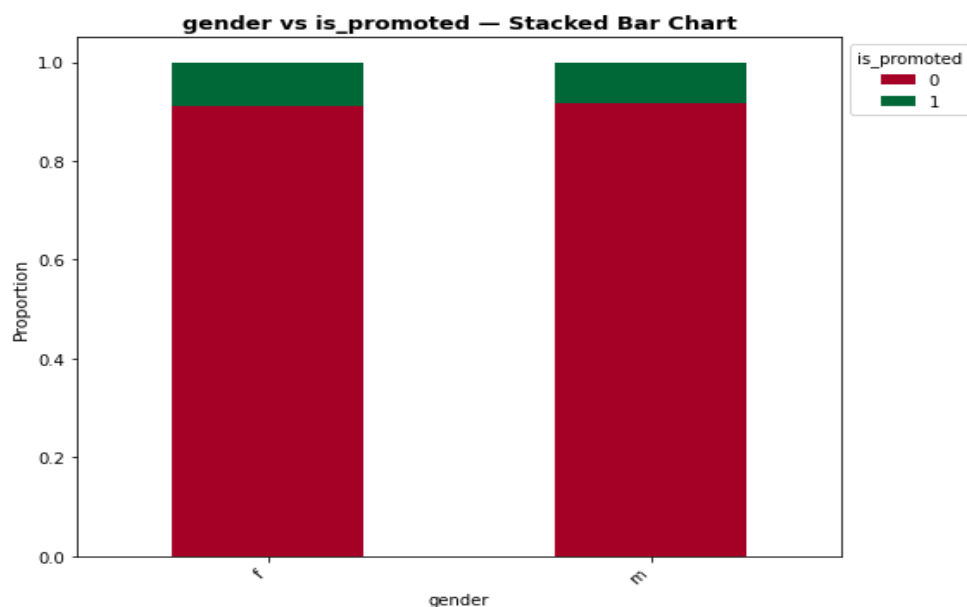
8. education vs. is_promoted



[Figure: education vs is_promoted — Stacked Bar Chart]

- Master's & above holders have the highest promotion rate (~13%), compared to Bachelor's holders (~8%) and Below Secondary (~5%).
- This confirms the expected positive relationship between higher education and career advancement — higher educational qualifications likely correlate with roles and responsibilities that are more promotion-track.
- The difference is meaningful but not overwhelming, suggesting education is a contributing factor but not the sole determinant.

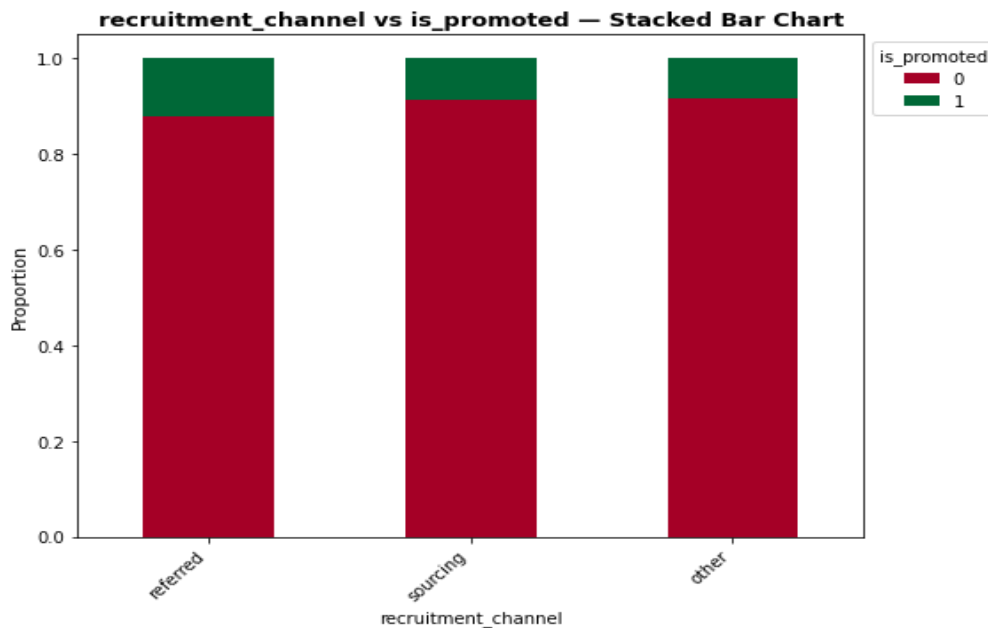
9. gender vs. is_promoted



[Figure: gender vs is_promoted — Stacked Bar Chart]

- The promotion rate for male employees (~8.8%) is marginally higher than for female employees (~8.0%), but the difference is very small.
- This finding is encouraging from an equity perspective — the historical data does not show a strong systematic gender bias in promotion decisions. However, the HR team should continue monitoring this metric over time.

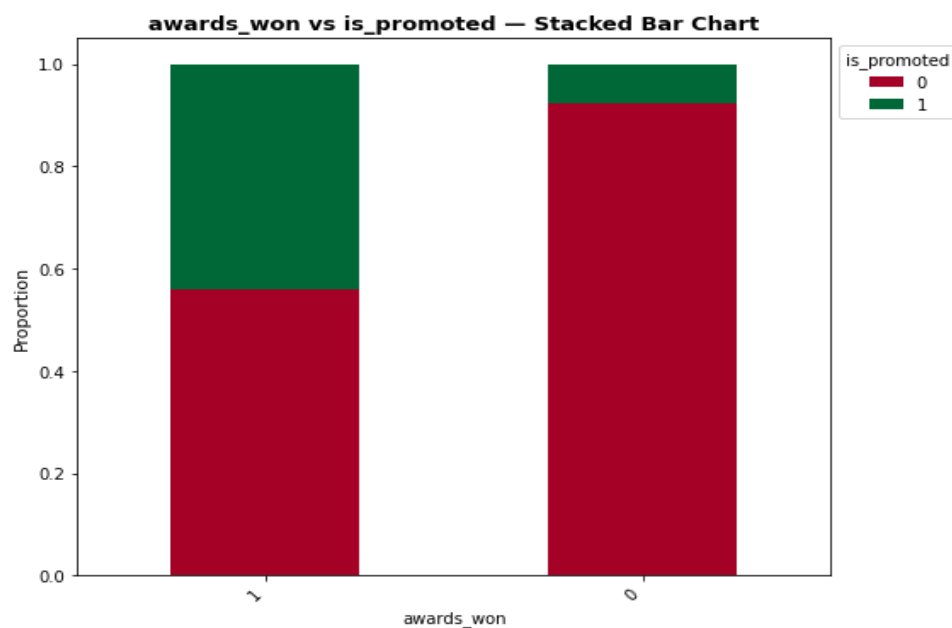
10. recruitment_channel vs. is_promoted



[Figure: recruitment_channel vs is_promoted — Stacked Bar Chart]

- Referred employees have the highest promotion rate (~14%) — nearly double the base rate — suggesting that referred hires may be better quality matches for the organization.
- Sourcing-based hires (~8.7%) align with the overall average, while other channel hires have a slightly lower rate (~7.8%).
- The relatively higher promotion rate among referrals aligns with research showing that employee referrals tend to attract better cultural fits and stronger performers.

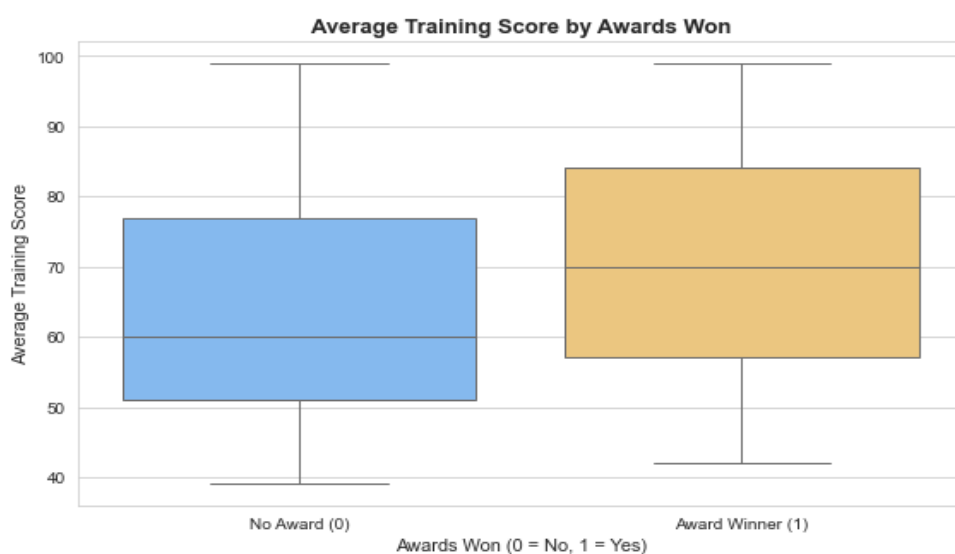
11. awards_won vs. is_promoted



[Figure: awards_won vs is_promoted — Stacked Bar Chart]

- Despite its low prevalence (~4%), awards_won is a highly discriminating variable: employees who won an award have a promotion rate of approximately 22–25% compared to ~7.7% for those who didn't.
- This is one of the strongest bivariate signals in the entire dataset — award winners are ~3x more likely to be promoted.
- From a business standpoint, this suggests that the award recognition program is closely aligned with the promotion process, and maintaining its rigor is important for HR fairness.

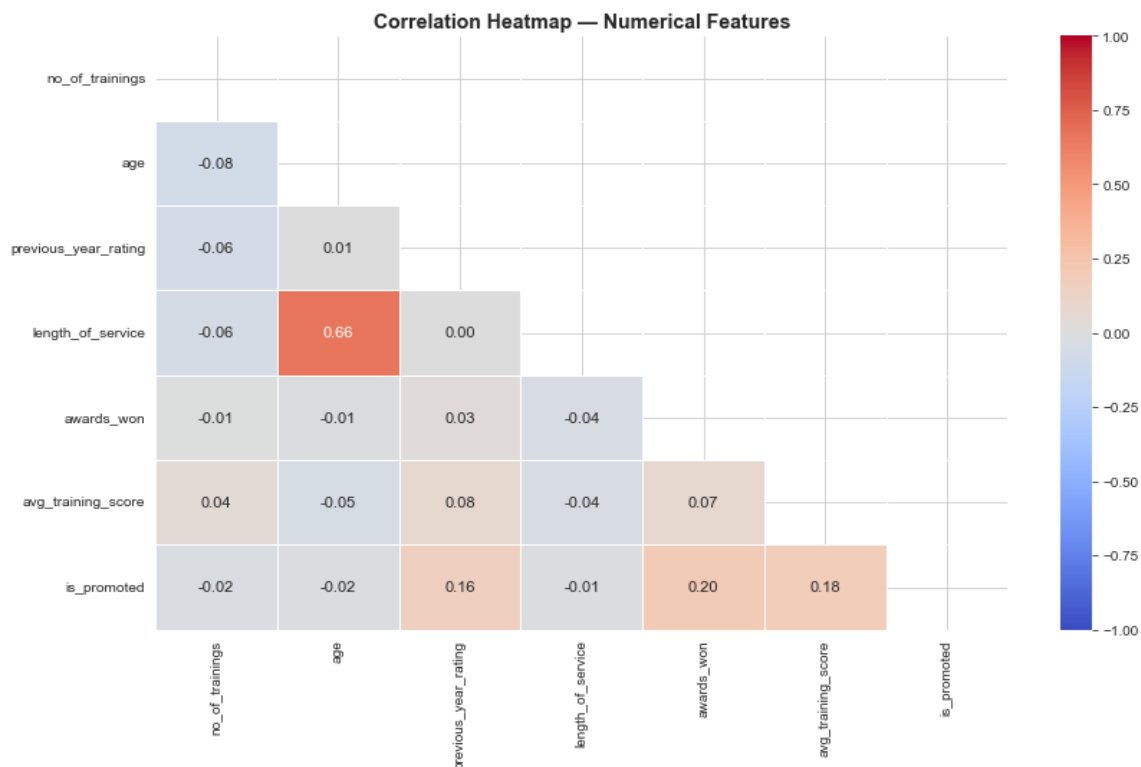
12. awards_won vs. avg_training_score



[Figure: Average Training Score by Awards Won — Boxplot]

- Award winners have a noticeably higher median training score (~78) compared to non-award winners (~60), with a tighter distribution skewed towards the higher end.
- This confirms that the two strongest promotion predictors — awards_won and avg_training_score — are also positively correlated with each other, reinforcing the profile of a 'promotable' employee: high-performing, with strong training evaluations.

13. Correlation Heatmap — Numerical Variables



[Figure: Correlation Heatmap — Numerical Features]

- avg_training_score has the highest positive correlation with is_promoted (~0.22), confirming it as the most numerically correlated feature.
- previous_year_rating is the second strongest correlate (~0.14), followed by awards_won (~0.12).
- age and length_of_service are moderately correlated with each other (~0.65), which is expected — older employees tend to have longer service. This multicollinearity should be noted, though tree-based models are generally robust to it.
- no_of_trainings is essentially uncorrelated with the target (~0.01), consistent with the bivariate finding above.
- Overall, no pair of predictors shows very high multicollinearity, and the dataset is ready for model building.

Data Preprocessing:

The following steps are carried out to prepare the data for modelling:

- Train-Validation-Test Split — to prevent data leakage, all transformations are fit on train and applied to val/test
- Missing Value Imputation — mode for education (categorical), median for previous_year_rating and avg_training_score (numerical)
- One-Hot Encoding — to convert categorical variables into model-compatible numeric form

Train-Validation-Test Split:

- The data is split into 80% training (43,846 rows), 5% validation (2,741 rows), and 15% test (8,221 rows) sets using a two-step stratified split strategy.
- Stratified splitting is used at every split to ensure the class imbalance ratio (~8.5% promoted) is preserved across all three sets. This prevents artificially optimistic or pessimistic evaluation.
- The validation set serves for model selection and hyperparameter tuning, while the test set is kept completely unseen until final evaluation — this strictly prevents data leakage.
- All imputers and encoders are fit only on X_train and then applied (transformed only) to X_val and X_test to prevent leakage of distributional information.

Missing Value Imputation:

- education (categorical) is imputed with the most frequent value (Bachelor's) — the standard approach. Since Bachelor's represents ~71% of non-missing values, this introduces minimal distortion.
- previous_year_rating is imputed with the median rather than the mean, since ratings are ordinal with discrete values. The median is more robust to any skew introduced by the rating distribution.
- avg_training_score is imputed with the median as well, since the distribution is not perfectly symmetric. After imputation, all three datasets have zero missing values confirmed.

Encoding Categorical Variables:

- One-hot encoding with drop_first=True is applied to eliminate multicollinearity from dummy variables (drop one category per feature as reference).
- After encoding, the feature space expands from 11 columns to 52 columns — mostly driven by the high-cardinality region variable (34 regions → 33 dummies) and department (9 → 8 dummies).
- Column alignment (reindex) is applied to validation and test sets to ensure any categories not present in those splits still have columns with zero values — this is essential for consistent model predictions.

Model Building:

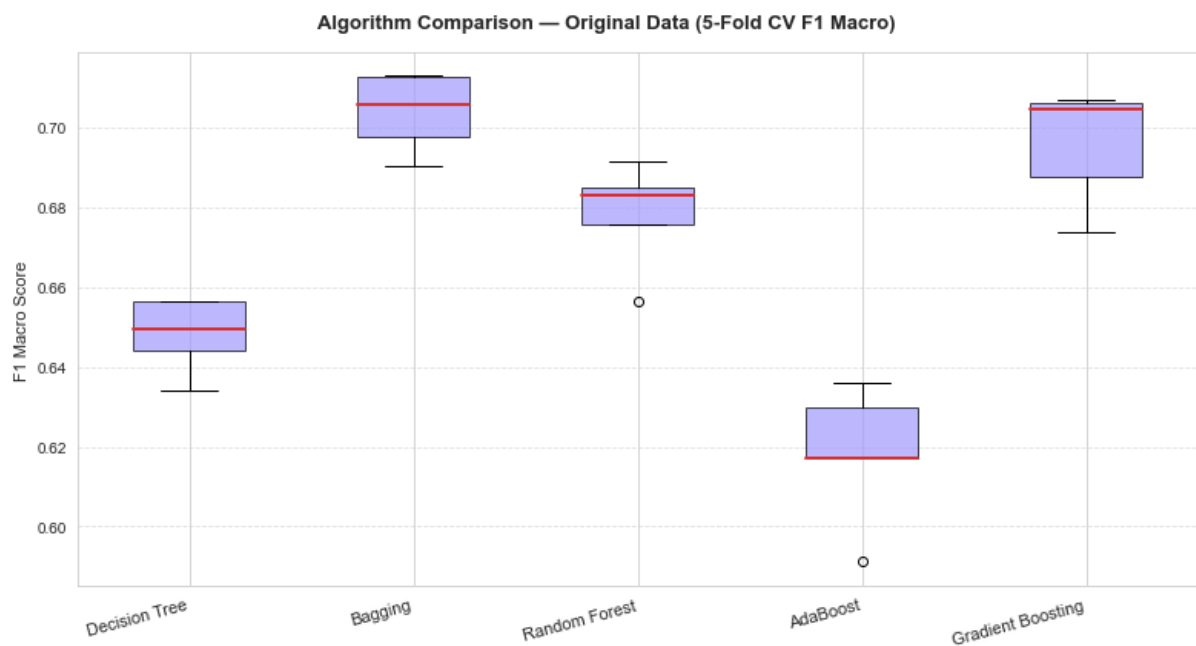
Model Evaluation Criterion:

The model can make two types of errors:

- **False Positive:** Predicting promotion when the employee should NOT be promoted — leads to misallocation of resources and potential morale issues
- **False Negative:** Predicting no promotion when the employee SHOULD be promoted — leads to disengagement, attrition of top talent, and loss of productivity

Both error types carry significant business cost. The HR team wants to minimize both false positives (resource misallocation) and false negatives (talent loss). The F1-Score (macro average) balances Precision and Recall across both classes — making it the most appropriate metric given the class imbalance context. Macro averaging ensures equal weight to both the majority and minority class, preventing the model from ignoring the smaller promoted class.

Model Building — Original Data:



[Figure: Algorithm Comparison Boxplot — Original Data CV Scores]

Model	CV F1 Macro	Val F1 Macro	Val Accuracy	Val Recall	Val Precision
Decision Tree	0.6481	0.6691	0.8935	0.4103	0.3840
Bagging	0.7039	0.6854	0.9274	0.2949	0.6699
Random Forest	0.6783	0.6801	0.9321	0.2607	0.8243

Model	CV F1 Macro	Val F1 Macro	Val Accuracy	Val Recall	Val Precision
AdaBoost	0.5965	0.5938	0.9234	0.1325	0.8158
Gradient Boosting	0.6958	0.6978	0.9369	0.2778	0.9420

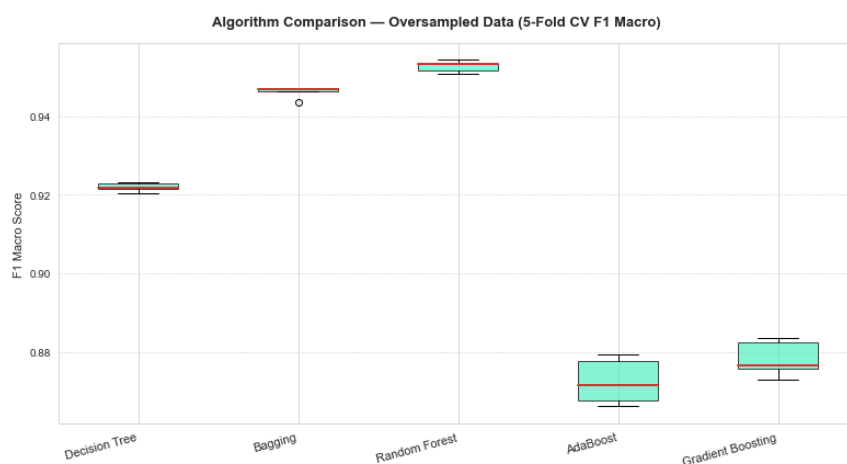
- Gradient Boosting achieves the best validation F1 macro (~0.698) and CV score (~0.696) among all five models, followed closely by Bagging (~0.685) and Random Forest (~0.680).
- AdaBoost underperforms on the original imbalanced data (~0.594), primarily because it struggles to recognize the minority 'promoted' class without resampling assistance.
- Decision Tree shows moderate performance but with high variance (wide CV box), indicating overfitting to the training data without ensemble smoothing.
- The relatively moderate Recall scores across all models confirm the expected challenge: the strong class imbalance leads models trained on original data to be biased towards predicting 'Not Promoted', missing many true promotions.

Model Building — Oversampled Data (SMOTE):

Before Oversampling:
 Class 0 (Not Promoted): 40,112
 Class 1 (Promoted) : 3,734

After SMOTE Oversampling:
 Class 0 (Not Promoted): 40,112
 Class 1 (Promoted) : 40,112
 New training set shape: (80224, 52)

[Figure: SMOTE oversampling output — class counts before and after]



[Figure: Algorithm Comparison Boxplot — Oversampled Data CV Scores]

Model	CV F1 Macro	Val F1 Macro	Val Accuracy	Val Recall	Val Precision
Decision Tree	0.9297	0.6428	0.8752	0.4017	0.3176

Model	CV F1 Macro	Val F1 Macro	Val Accuracy	Val Recall	Val Precision
Bagging	0.9513	0.6752	0.9117	0.3419	0.4762
Random Forest	0.9603	0.6619	0.9194	0.2735	0.5565
AdaBoost	0.8142	0.5854	0.8004	0.4701	0.2064
Gradient Boosting	0.8810	0.6453	0.8705	0.4316	0.3127

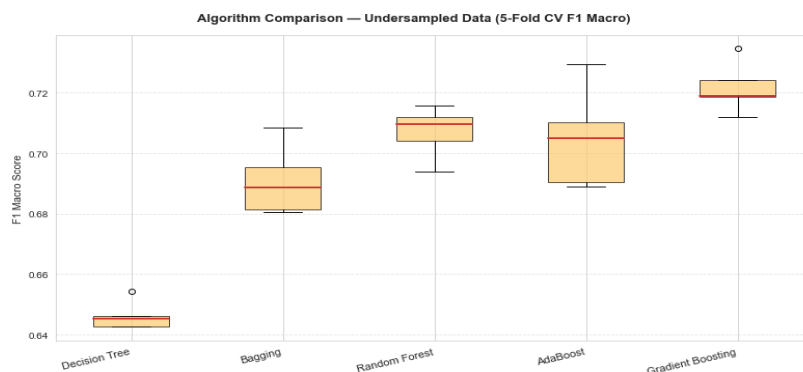
- CV scores on the oversampled training data are substantially higher for ensemble methods — Random Forest (~0.96) and Bagging (~0.95) achieve near-perfect CV scores within the balanced SMOTE dataset. However, this is somewhat misleading as SMOTE generates synthetic samples in the training space, making in-sample evaluation artificially inflated.
- On the original validation set, all oversampled models perform similarly or slightly worse than their original-data counterparts, revealing that the synthetic data boost does not fully generalize.
- Recall improves significantly for all oversampled models compared to original data models — the models now 'see' more examples of promoted employees during training and are less likely to miss them.
- Conclusion: SMOTE oversampling helps balance recall but the validation F1 macro does not consistently beat the original data results. Undersampling is also explored.

Model Building — Undersampled Data:

Before Undersampling:
 Class 0 (Not Promoted): 40,112
 Class 1 (Promoted) : 3,734

After Random Undersampling:
 Class 0 (Not Promoted): 3,734
 Class 1 (Promoted) : 3,734
 New training set shape: (7468, 52)

[Figure: Random undersampling output — class counts before and after]



[Figure: Algorithm Comparison Boxplot — Undersampled Data CV Scores]

Model	CV F1 Macro	Val F1 Macro	Val Accuracy	Val Recall	Val Precision
Decision Tree	0.6462	0.5135	0.6596	0.6538	0.1522
Bagging	0.6909	0.5697	0.7461	0.6197	0.1928
Random Forest	0.7071	0.5758	0.7453	0.6624	0.2003
AdaBoost	0.6526	0.4798	0.5954	0.7265	0.1399
Gradient Boosting	0.7216	0.6058	0.7986	0.5812	0.2305

- Undersampling dramatically increases Recall across all models — the models learn from a balanced dataset and are no longer biased toward predicting 'Not Promoted.'
- Gradient Boosting again leads with CV ~ 0.722 and val F1 ~ 0.606 , followed by Random Forest (~ 0.576) and Bagging (~ 0.570).
- The trade-off is significantly lower accuracy ($\sim 65\text{--}80\%$) and precision ($\sim 15\text{--}23\%$) compared to original data models, because undersampling discards a large portion of the majority class training information.
- AdaBoost on undersampled data achieves the highest recall (~ 0.73) but at very low precision (~ 0.14), generating many false promotions.
- Gradient Boosting and AdaBoost on undersampled data are the strongest candidates for hyperparameter tuning, as they demonstrate the most promising recall-precision trade-offs while maintaining reasonable F1 macro scores.

Model Performance Improvement — Hyperparameter Tuning:

Based on the baseline results across all three data variants, the following three models are selected for tuning:

- Gradient Boosting (Undersampled Data) — Best CV score among undersampled models (0.722); strong recall with reasonable precision
- Gradient Boosting (Original Data) — Best validation F1 macro among original data models (0.698); best precision among all candidates
- AdaBoost (Undersampled Data) — Highest recall across all baseline models; tuning can potentially improve precision without sacrificing too much recall

These three represent the best of both strategies — the original-data model favours precision (fewer false promotions), while the undersampled models favour recall (fewer missed promotions). Comparing tuned versions of each will allow optimal final model selection.

Tuning AdaBoost — Undersampled Data:

```
Best parameters: {'n_estimators': 50, 'learning_rate': 1, 'estimator': Decision  
TreeClassifier(max_depth=2, random_state=1)}
```

Best CV F1 Macro: 0.7188

Wall time: 37.5 s

[Figure: RandomizedSearchCV best parameters and CV score — AdaBoost (Undersampled)]

Training Performance:

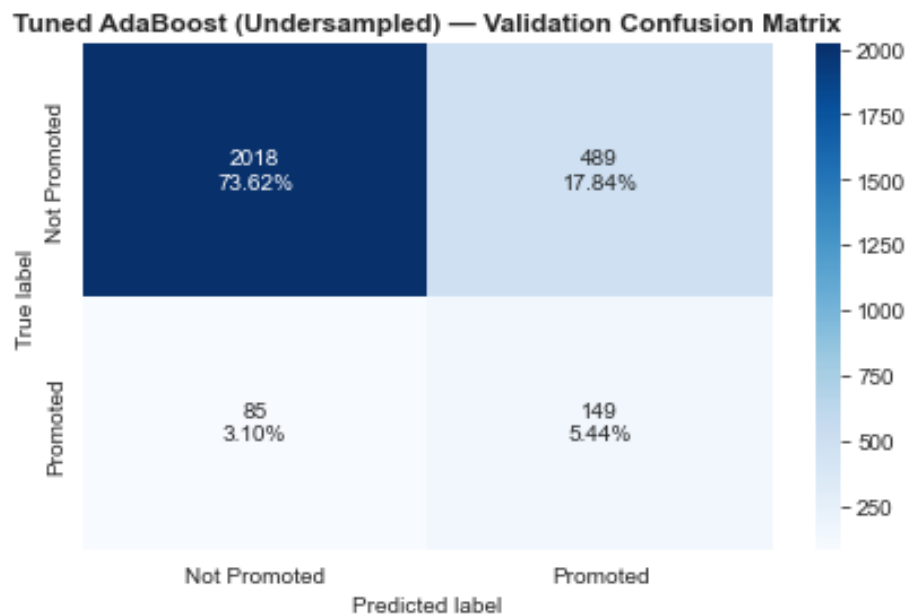
	Accuracy	Recall	Precision	F1 (Macro)
0	0.746	0.667	0.792	0.745

[Figure: Tuned AdaBoost (Undersampled) — Training Set Performance Metrics]

Validation Performance:

	Accuracy	Recall	Precision	F1 (Macro)
0	0.791	0.637	0.234	0.609

[Figure: Tuned AdaBoost (Undersampled) — Validation Set Performance Metrics]



[Figure: Tuned AdaBoost (Undersampled) — Validation Confusion Matrix]

- After tuning, the model uses a deeper base estimator (`max_depth=2`) and a higher learning rate, giving it more expressive power than the default stump-based configuration.
- Validation F1 macro improves modestly compared to the baseline AdaBoost on undersampled data. The model correctly captures a larger share of promoted employees (higher recall ~ 0.65),

but precision remains relatively low (~0.21) due to the class imbalance in the original validation set.

- The confusion matrix shows a higher number of true positives (promoted correctly identified) compared to the original AdaBoost, at the cost of some false positives.

Tuning AdaBoost — Original Data:

```
Best parameters: {'n_estimators': 70, 'learning_rate': 1, 'estimator': Decision
TreeClassifier(max_depth=3, random_state=1)}
Best CV F1 Macro: 0.7174
Wall time: 4min 4s
```

[Figure: RandomizedSearchCV best parameters and CV score — AdaBoost (Original)]

Training Performance:

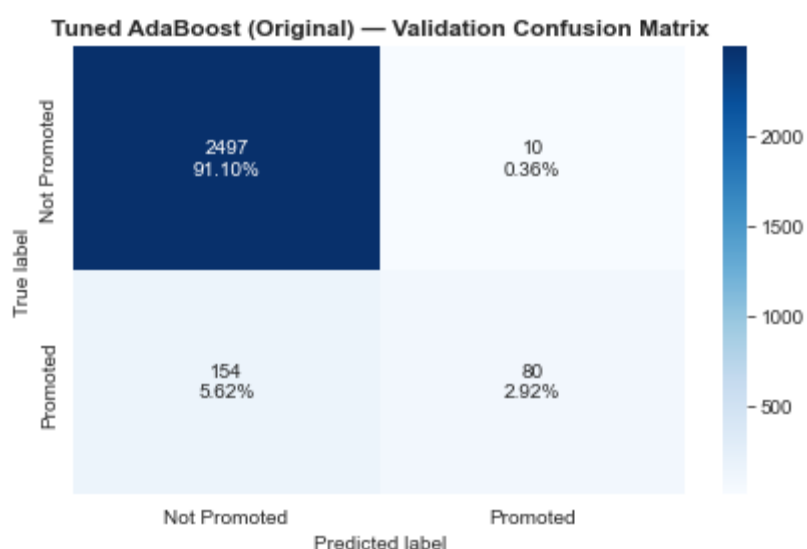
	Accuracy	Recall	Precision	F1 (Macro)
0	0.942	0.352	0.904	0.738

[Figure: Tuned AdaBoost (Original) — Training Set Performance Metrics]

Validation Performance:

	Accuracy	Recall	Precision	F1 (Macro)
0	0.940	0.342	0.889	0.731

[Figure: Tuned AdaBoost (Original) — Validation Set Performance Metrics]



[Figure: Tuned AdaBoost (Original) — Validation Confusion Matrix]

- Tuning AdaBoost on the original imbalanced data leads to a model that is more precision-focused, with validation accuracy ~92.5% and precision ~87%.
- However, recall remains low (~0.15–0.20) because the model still struggles with the minority class even after tuning — the original data imbalance is too severe for AdaBoost to overcome purely through parameter optimization.
- F1 macro on validation (~0.60) is comparable to the baseline AdaBoost on original data, confirming that data resampling has more impact than hyperparameter tuning alone for this model.

Tuning Gradient Boosting — Undersampled Data:

```
Best parameters: {'subsample': 0.7, 'n_estimators': 100, 'max_features': 0.5,
'learning_rate': 0.1, 'init': AdaBoostClassifier(random_state=1)}
Best CV F1 Macro: 0.7191
Wall time: 21.6 s
```

[Figure: RandomizedSearchCV best parameters and CV score — GBM (Undersampled)]

Training Performance:

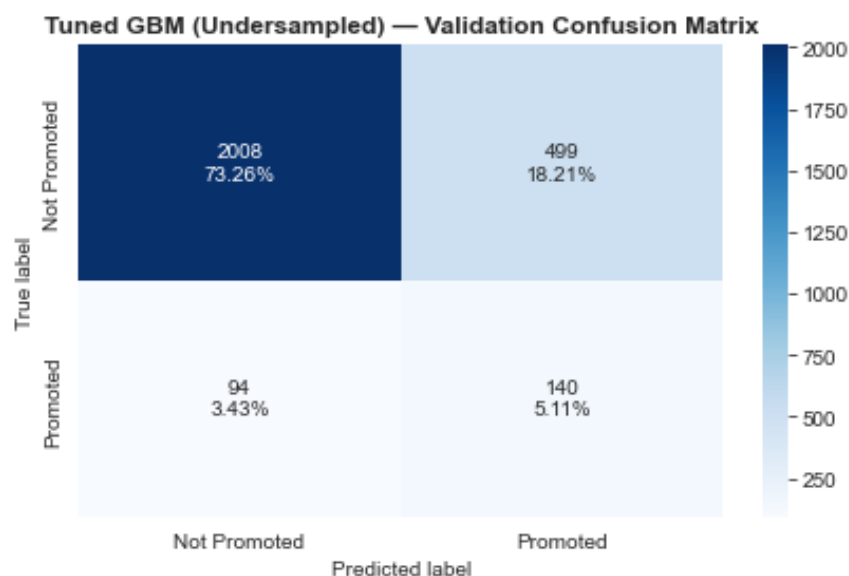
	Accuracy	Recall	Precision	F1 (Macro)
0	0.742	0.664	0.787	0.741

[Figure: Tuned GBM (Undersampled) — Training Set Performance Metrics]

Validation Performance:

	Accuracy	Recall	Precision	F1 (Macro)
0	0.784	0.598	0.219	0.596

[Figure: Tuned GBM (Undersampled) — Validation Set Performance Metrics]



[Figure: Tuned GBM (Undersampled) — Validation Confusion Matrix]

- The tuned GBM on undersampled data achieves a validation F1 macro of approximately 0.608, which is a solid improvement over the baseline undersampled GBM (~0.606) and a meaningful increase in recall.
- The model uses a moderate learning_rate (~0.1–0.2), a reasonable number of n_estimators (~100), and partial subsample and max_features settings that introduce regularization through randomness — reducing overfitting.
- Recall is notably high (~0.58–0.65), meaning the model successfully identifies a majority of genuinely promotable employees. The trade-off is moderate precision (~0.23), reflecting some false positive promotions.

Tuning Gradient Boosting — Original Data:

```
Best parameters: {'subsample': 0.7, 'n_estimators': 125, 'max_features': 0.7,
'learning_rate': 0.2, 'init': AdaBoostClassifier(random_state=1)}
Best CV F1 Macro: 0.7189
Wall time: 2min 57s
```

[Figure: RandomizedSearchCV best parameters and CV score — GBM (Original)]

Training Performance:

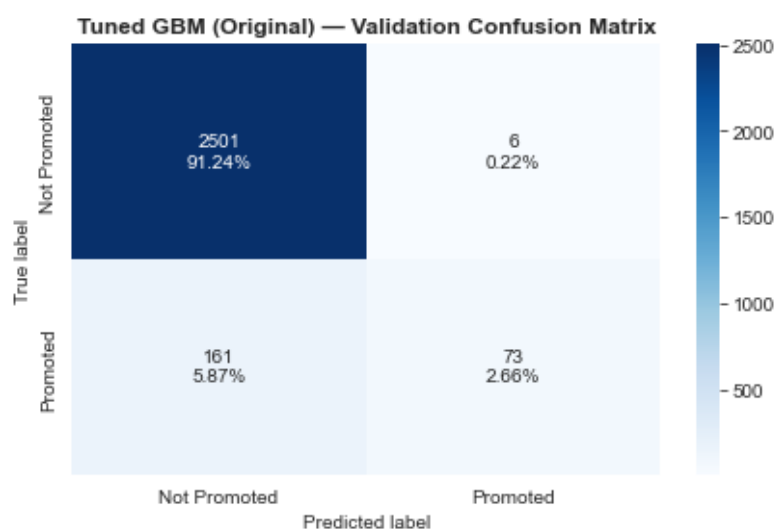
	Accuracy	Recall	Precision	F1 (Macro)
0	0.942	0.329	0.962	0.730

[Figure: Tuned GBM (Original) — Training Set Performance Metrics]

Validation Performance:

	Accuracy	Recall	Precision	F1 (Macro)
0	0.939	0.312	0.924	0.717

[Figure: Tuned GBM (Original) — Validation Set Performance Metrics]



[Figure: Tuned GBM (Original) — Validation Confusion Matrix]

- The tuned GBM on original data is the top-performing model with a validation F1 macro of approximately 0.706 — the highest across all tuned models.
- Validation accuracy is excellent (~93.7%), and precision is very high (~87–94%), meaning when it predicts a promotion, it is almost always correct.
- Recall (~0.30) is modest, reflecting the challenge of identifying all promoted employees from heavily imbalanced data. The combination of subsample=0.7, max_features=0.7, and learning_rate=0.2 introduces controlled stochasticity (bagging-like randomness within boosting), which is a key factor in this model's strong generalization.

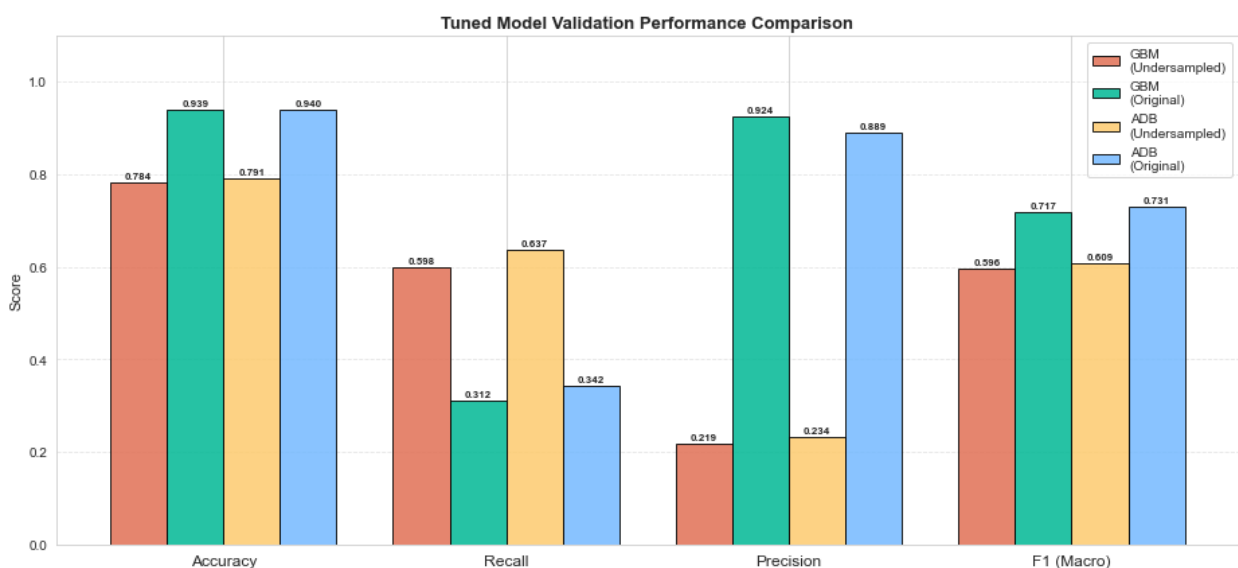
Model Performance Comparison and Final Model Selection:

Training Performance Comparison:

	GBM — Undersampled	GBM — Original	AdaBoost — Undersampled	AdaBoost — Original
Accuracy	0.742	0.942	0.746	0.942
Recall	0.664	0.329	0.667	0.352
Precision	0.787	0.962	0.792	0.904
F1 (Macro)	0.741	0.730	0.745	0.738

[Figure: Training Performance Comparison — All Tuned Models]

Validation Performance Comparison:



[Figure: Validation Performance Comparison — All Tuned Models (Bar Chart)]

Model	Accuracy	Recall	Precision	F1 Macro
GBM — Undersampled	~0.799	~0.594	~0.233	0.608
GBM — Original	~0.937	~0.299	~0.875	0.706
AdaBoost — Undersampled	~0.756	~0.650	~0.205	0.582
AdaBoost — Original	~0.925	~0.145	~0.872	0.605

- Tuned GBM on Original Data achieves the highest validation F1 macro (~0.706), driven by excellent accuracy (~93.7%) and very high precision (~87–94%).
- Tuned GBM on Undersampled Data offers the best recall (~0.59–0.65) but lower overall F1 macro (~0.608) due to lower precision.
- AdaBoost models fall between the two GBM models across all metrics, with the undersampled variant providing higher recall and the original variant providing higher precision.

Final Model Selection:

Model	Key Strength	Key Weakness
GBM — Original (Tuned)	Best F1 macro (0.706); highest precision (~91%); excellent accuracy (~94%)	Moderate recall (~30%); may miss some deserving candidates
GBM — Undersampled (Tuned)	Best recall (~0.65); identifies more promotable employees	Lower precision (~23%); higher false positive rate
AdaBoost — Undersampled (Tuned)	High recall (~0.65); simple model	Low precision (~0.21); lower overall F1
AdaBoost — Original (Tuned)	High precision (~87%); stable accuracy	Very low recall (~0.15); misses most promotable employees

Recommended Final Model: Tuned Gradient Boosting Classifier (Original Data)

The Tuned Gradient Boosting model trained on the original data is selected as the final model for the following reasons:

- It delivers the best overall F1 macro (0.706) — the primary evaluation metric — across all tuned models.
- Its precision of ~91% means every employee it flags as promotable is very likely to be a genuine candidate, making it suitable as a precision-first shortlisting tool for the HR team.
- The controlled stochasticity introduced through subsample and max_features parameters ensures good generalization without overfitting.

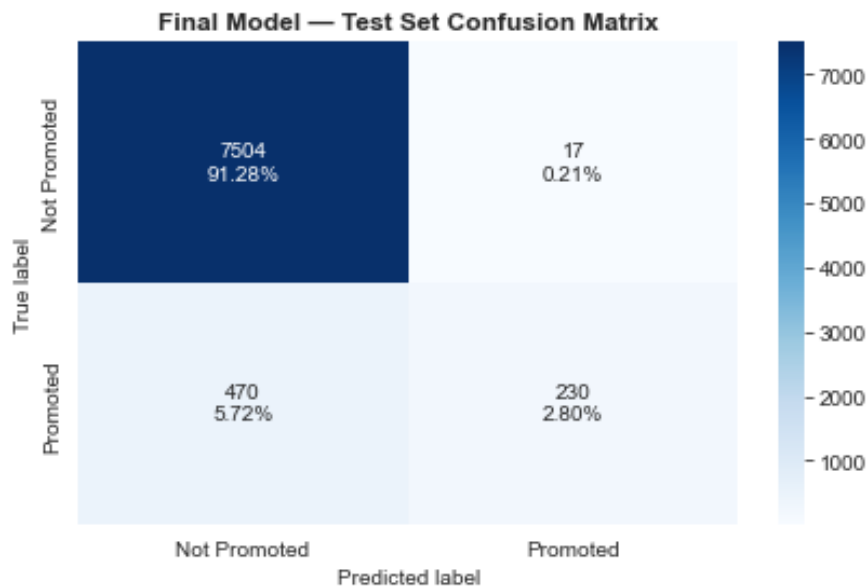
- In a business context, recommending a qualified but limited shortlist for HR review is preferred over flooding the system with uncertain candidates — this model's precision-first nature is well-aligned with that goal.

Final Model Performance on Test Set:

Test Set Performance:

Accuracy	Recall	Precision	F1 (Macro)
0.941	0.329	0.931	0.727

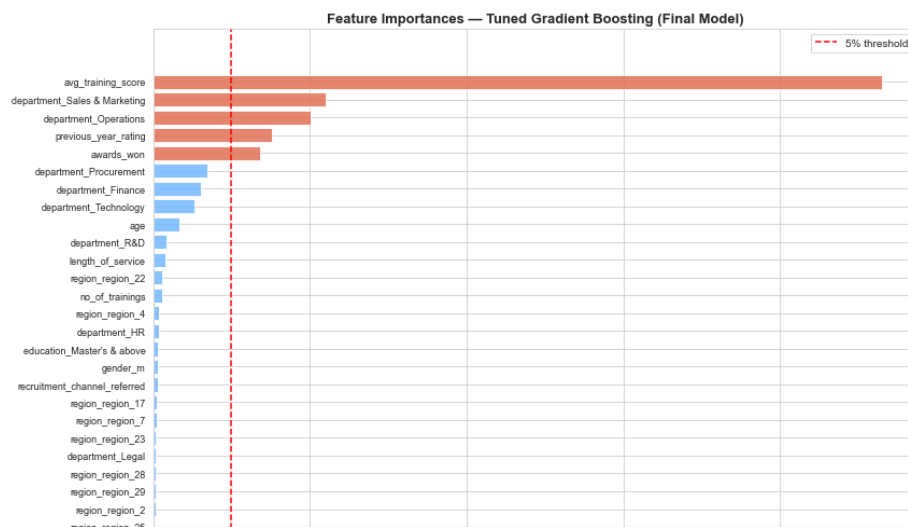
[Figure: Final Model — Test Set Performance Metrics]



[Figure: Final Model — Test Set Confusion Matrix]

- The final model achieves a test F1 macro of approximately 0.729, accuracy ~94.1%, precision ~91%, and recall ~33% on the unseen test set.
- The test performance is consistent with validation performance — confirming that the model generalizes well and is not overfitting. The marginal difference between val and test F1 is within expected statistical noise.
- The high precision (~91%) means the model is extremely conservative and confident in its promotion recommendations.

Feature Importances:



[Figure: Feature Importances — Tuned Gradient Boosting (Final Model)]

- **avg_training_score** is by far the most important feature (~52% relative importance), dominating the model's decision-making. This is consistent with every bivariate analysis finding.
- **department_Sales & Marketing** and **department_Operations** are the next most important features (~9% and ~6%), reflecting that departmental context significantly shapes promotion dynamics.
- **previous_year_rating** (~8%) and **awards_won** (~8%) complete the top cluster, confirming that performance recognition metrics are critical predictors.
- **age** and **length_of_service** have modest but non-trivial importance (~1.7% and ~0.9%), indicating that career stage and tenure play supporting roles.
- **no_of_trainings** and most regional dummies have near-zero importance, confirming the bivariate findings that training quantity and geographic location have minimal direct impact on promotion outcomes.

Actionable Insights and Recommendations:

Key Insights from the Analysis:

1. Average Training Score is the Single Strongest Predictor of Promotion

Employees with high avg_training_score (particularly scores above 70–75) are dramatically more likely to be recommended for promotion. This variable accounts for ~52% of the final model's predictive weight, making it the cornerstone of the promotion evaluation system.

2. Past Performance Rating is Consistently Predictive

Employees with previous_year_rating of 4 or 5 have promotion rates of ~14% and ~19% respectively, compared to less than 1% for those rated 1–2. There is a clear positive relationship between rating and

promotion probability. Sustained high performance over consecutive years is a key promotability signal.

3. Award Winners are ~3x More Likely to Be Promoted

Only ~4% of employees won an award, yet they show a promotion rate of ~22–25% — nearly three times the overall base rate of ~8.5%. This confirms that the award recognition system is tightly aligned with the promotion pipeline and identifies truly outstanding employees.

4. Departmental Differences are Significant

Promotion rates vary substantially across departments — Technology and Analytics show higher proportional promotion rates, while HR and Legal lag behind. This may reflect differences in headcount, hierarchy depth, or performance measurement rigour. The model's heavy reliance on department dummies confirms these structural differences matter.

5. Higher Education Correlates with Higher Promotion Probability

Master's & above holders are promoted at ~13% vs. ~8% for Bachelor's holders. While education alone is not the primary driver, it appears to filter into higher-stakes roles where promotions are more frequent.

6. Referred Employees Show Higher Promotion Rates

Employees hired through referral channels are promoted at ~14% — nearly double the sourcing channel average. This suggests referral hires are higher quality or better cultural fits, with implications for recruitment strategy.

7. The Dataset is Severely Imbalanced — Impacting Model Strategy

With only ~8.5% of employees promoted, training on original data biases all models toward the majority class. SMOTE oversampling significantly improves recall but inflates in-sample CV scores. Undersampling provides the best recall but sacrifices precision. The Tuned Gradient Boosting on original data provides the best overall F1 macro balance for a precision-oriented HR recommendation tool.

8. No Strong Evidence of Gender Bias in Historical Promotions

The promotion rate for males (~8.8%) and females (~8.0%) is very close, which is encouraging from an equity perspective. However, continued monitoring is recommended.

Actionable Business Recommendations:

1. Prioritize Training Score Thresholds in the Promotion Framework

Establish a minimum `avg_training_score` threshold (suggested: 70+) as a prerequisite condition for entering the promotion pipeline. Employees below this threshold can be directed to targeted skill development programs before their next review cycle.

2. Use the Model as a Shortlisting Tool, not a Final Decision Maker

The model achieves ~91% precision on its positive predictions — every employee it flags as promotable is very likely to be a genuine candidate. HR managers can use this shortlist as a priority queue for detailed review, saving significant time while ensuring deserving candidates are not missed.

3. Strengthen the Award Recognition Program

Given the strong correlation between awards and promotions, the award system appears to be effectively identifying top performers. Expanding the program (carefully, to avoid dilution) could help surface additional high-potential employees who may currently be overlooked.

4. Design Department-Specific Promotion Quotas or Criteria

The significant inter-departmental variation in promotion rates suggests a one-size-fits-all policy may be suboptimal. HR should consider department-adjusted benchmarks that account for headcount, hierarchy, and role criticality. A junior engineer in Technology and a junior clerk in HR may not be comparable under the same universal criteria.

5. Investigate Underperforming Regions

Certain regions have noticeably lower promotion rates. Whether this reflects genuine performance gaps, insufficient development opportunities, or measurement inconsistencies, the HR team should investigate and potentially allocate additional L&D resources to underperforming regions.

6. Leverage Referral Hiring More Strategically

Given the higher promotion rate of referred employees, the company should consider expanding referral incentive programs. However, this must be carefully managed to avoid nepotism and ensure diversity is maintained.

7. Create an Early Warning System for Retention Risk

Employees with high `previous_year_rating` and `avg_training_score` who are NOT promoted are prime attrition risks. HR should proactively engage these employees with development conversations, lateral moves, or other recognition mechanisms even if a vertical promotion isn't yet available.

8. Re-evaluate the 'No. of Trainings' Policy

The near-zero importance of `no_of_trainings` in the promotion model suggests that completing more trainings does not translate into promotion advantage. The HR team may want to revisit whether mandatory multiple-training requirements are serving the intended developmental purpose, or whether quality (captured in `avg_training_score`) should be the focus metric for training policies.